Study of bacteria-substrate interactions for improved antibacterial material design

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Abstract - *Staphylococcus aureus (S. Aureus)* is one of the major pathogenic causes of implant failure and post-operative infections. The probability of a deep infection post a total knee arthroplasty is 20%. A study by Mandya Institute of Medical Sciences revealed that primary cause of infections was *S. aureus* bacteria in 55% cases. The interaction between *S. aureus* and silica (implant surface), which also leads to biofilm formation is crucial for understanding the infections on medical implants and designing anti-bacterial substrates. This study involves extensive use of NAMD (Nano-scale Molecular Dynamics) and VMD (Visual Molecular Dynamics) for employing the concept of molecular dynamics on bacteria-substrate interaction. It models the silica substrate surface and captures the unfolding of three adhesins of *S. aureus* (SpA, ClfA, and SraP), in a hydrated, ionic environment which simulates real-life physiological conditions. The outputs of NAMD are compatible with VMD for visualization and, CHARMM and X-PLOR for applying the force field over the simulation. Further NAMD simulations like SMD (Steered Molecular Dynamics) quantifies the adhesion forces by applying controlled external forces to specific atoms or domains. This computational approach paves the way for better designing of bio-glass based implants, with better resistance to bio-film formation and improved post-operation patient safety.

***Keywords*:** *adhesion forces, silica, molecular dynamics, Staphylococcus aureus, ubiquitin*

1 INTRODUCTION

A study from a teaching hospital in India reported that *Staphylococcus aureus* was the most common causative microorganism in orthopedic implant-associated infections, responsible for 33% of culture-positive samples. Of these, 60% were methicillin-resistant Staphylococcus aureus (MRSA) strains[1]. Silica is widely used in dental and orthopedic implants, to improve their bioactivity and integration with tissue, making it a practical and translationally relevant model for studying bacterial-substrate interactions in the biomedical field[2]. *S. aureus* can easily adhere to implant surfaces and initiate the formation of biofilms which are complex, organized bacterial communities embedded within a self-produced extracellular matrix[3]. This matrix acts as a protective barrier, impeding the effectiveness of antibiotics and shielding bacteria from the host immune response, which makes such infections extremely challenging to eliminate[4]. Since bacterial adhesion is the critical first step in biofilm development, it plays a key role in the persistence and resilience of implant-associated infections[5],[6]. By simulating and quantitatively analyzing how *S. aureus* attaches to silica surfaces, valuable insights for engineering implant materials that deter bacterial colonization can be gained.

The primary obstacles to this aim, however, arise from the intricate biology of *S. aureus*, which complicates the accurate modeling of all factors that may influence adhesion[7]. Moreover, the limited availability of detailed reference data makes NAMD simulations of this bacterium highly resource-intensive. To address these challenges, it is necessary to first conduct preliminary studies on a simpler model system to gain foundational insights into the host cell's protein regulation mechanisms[8]. Ubiquitin, a protein found in nearly all eukaryotic cells and involved in a diverse range of cellular activities, is an ideal candidate for such investigations into protein regulation[9].

The main objective was to investigate how the protein behaves and maintains its structure under biologically meaningful conditions[10]. To accurately reflect the cellular environment, the protein must be surrounded by water via the process of solvation. In these simulations, ubiquitin was placed within a water sphere to represent a minimally solvated, non-periodic setting, as well as in a water box to better emulate the extensive aqueous environment typical of cells. The water box setup enables the use of advanced techniques like Particle Mesh Ewald (PME) for realistic modeling of long-range electrostatic interactions. Additionally, a generalized Born implicit solvent (GBIS) approach was employed, which approximates the influence of water without explicitly simulating each water molecule, thereby greatly reducing computational requirements[11].

NAMD was chosen as the Molecular Dynamics (MD) simulation platform due to its ability to efficiently manage large biomolecular systems. Its support for multi-core processing enables simulations to be completed more rapidly compared to other MD engines like AMBER or GROMACS[12]. NAMD is also versatile, accommodating both explicit solvent models (such as water spheres and water boxes) and implicit models (like the generalized Born approach), while incorporating sophisticated methods for calculating electrostatic interactions, including Particle Mesh Ewald (PME). All simulation trajectories were visualized using VMD. Key preparatory steps—such as structure setup, solvation, equilibration, energy conservation, and mechanical testing—were first carried out on ubiquitin. The close alignment of statistical measures, including RMSD, energy distributions, and force curves, confirmed the accuracy and reliability of the simulation system. This provided a strong foundation for subsequent simulations of *Staphylococcus aureus* on silica.

Atomistic models of the silica surface needed to be modelled using force fields that accurately depict chemical bonding. The silica supercell surface was generated using pymatgen, a versatile Python toolkit for materials analysis and structural manipulation, to produce a surface slab with a chosen Miller index, ensuring the correct crystallographic orientation for further simulations[13]. The Atomic Simulation Environment (ASE) library was then employed to modify the atomic configuration further by adding vacuum layers and adjusting the slab's size and dimensions.

These preliminary simulations confirmed the robustness of the computational framework and helped define key procedures for preparing, analyzing, and interpreting the systems.

2 Materials and Methods

All molecular dynamics simulations were conducted with NAMD 3.0.1 (`Linux-x86_64-multicore-CUDA`), enabling effective parallel processing on GPU clusters[14]. The simulations employed CUDA version 12.8 on NVIDIA GPUs to enhance extensive biomolecular computations. CHARMM27 was selected as the force field because of its suitability for protein systems and its integration of CMAP corrections, which improveed the precision of backbone dihedral angles and overall protein behavior. VMD 1.9.3 was utilized for visualization and trajectory analysis, offering resources for molecular rendering and post-processing of simulation data[15].

2.1 System Setup of Ubiquitin

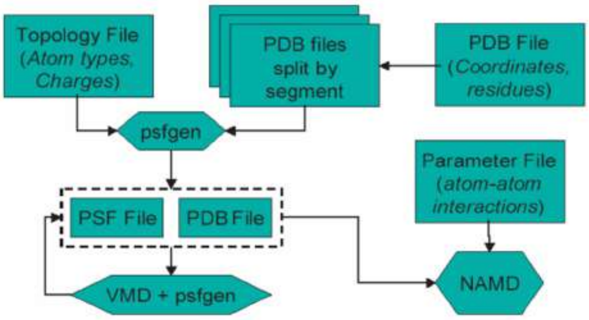


Figure 1: Flowchart indicating the role of files as used by VMD, NAMD, and psfgen.

The original ubiquitin structure file, `1UBQ.pdb`, was obtained from the Protein Data Bank. The simulation employed the CHARMM27 parameter and topology files, specifically `par_all27_prot_lipid.inp` and `top_all27_prot_lipid.inp`, which supplied force field information tailored for proteins and lipids. The psfgen module in VMD was utilized to create the appropriate PSF file for ubiquitin. A script file, `ubq.pgn`, was generated to guide psfgen in constructing the topology and deducing the coordinates of any absent atoms (like hydrogen atoms) based on the definitions specified in the topology file.

2.2 Solvating Ubiquitin

The produced PSF file included hydrogen atoms but was missing solvation, requiring the addition of explicit water molecules to replicate physiological conditions. To address both non-periodic and periodic boundary conditions, two solvation methods were utilized: a spherical water shell and a cubic water box.

A water sphere model was generated using the `wat_sphere.tcl` script in VMD for non-periodic simulations. This enclosed the protein within a droplet of TIP3P water, with the sphere's center and radius calculated automatically[17]. Spherical boundary conditions and electrostatics were addressed through the Multilevel Summation Method (MSM) as utilized in NAMD.

For periodic simulations, the protein was solvated in a cubic water box with VMD's solvate plugin, featuring a consistent padding of 5 Å around the protein in every direction. This periodic configuration allowed for the implementation of Particle Mesh Ewald (PME) for precise long-range electrostatic computations with isotropic pressure coupling. The origin for defining the periodic cell dimensions in the simulation input was established using the `measure center` command in VMD at the center of the solvation box (refer to Figure 2(b)).

2.3 Minimization and Equilibration of the Ubiquitin-Water Setup

Minimization of the protein structure was done in order to ensure the protein was in a local minimum on the potential energy surface, free from steric or atomic clashes which hence gave a physically stable basis to the net forces on the protein before proceeding with the simulations[18]. The water box and the water sphere systems were subjected to 100 steps of energy minimization and all atomic velocities were set to zero prior to the equilibration step. The system was then gradually heated to the desired temperature (310 K) and brought to thermodynamic equilibrium at the same, using a controlled Langevin thermostat[19]. Each equilibration simulation goes for 2500 steps (5 ps), and trajectory data were saved for every 100 steps. The same was done with a Generalized Born Implicit Solvent (GBIS). An implicit solvent model is a simulation technique which eliminates the need for explicit water atoms by including many of the effects of solvent in the inter-atomic force calculation. The configuration file taking the above details into account was run on NAMD.

2.4 Root Mean Square Deviation (RMSD) of Ubiquitin residues

Per-residue RMSD evaluation was performed with VMD on the equilibrated trajectory file `ubq_ws_eq.dcd` and structure file `ubq_ws.psf`, sourced from water sphere equilibration at 310 K[20]. The RMSD calculation included only backbone atoms, including N, Cα, and C atoms. The trajectory included 2500 frames recorded every 2 fs across 5 fs, produced from the `ubq_ws_eq.conf` setup. Water molecules were omitted from the selection, and solely protein residues were examined. A 12 Å cutoff, 10 Å switching distance, and PME with approximately 1.0Å grid spacing were employed during trajectory generation.

2.5 Cooling Behavior of Ubiquitin in a Water Sphere

Cooling simulations were conducted in NAMD, utilizing the CHARMM27 force field to examine thermal relaxation of ubiquitin within a TIP3P water sphere. The system was set up using `ubq_ws.psf` and `ubq_ws.pdb`, with initial coordinates and velocities extracted from the last frame of the equilibrated trajectory `ubq_ws_eq.dcd`. Simulations were executed with non-periodic boundary conditions employing a spherical harmonic potential specified by the `wat_sphere.tcl` output. The temperature was decreased linearly from 310 K to 100 K over 2000 steps utilizing Langevin dynamics with a damping constant of 5 ps^{-1}. A timestep of 2 fs was utilized, and all bonds that included hydrogen atoms were restricted. Electrostatics were analyzed utilizing the Multilevel Summation Method (MSM). Temperature and total energy were measured every 20 steps, and the temperature profile was graphed to verify cooling characteristics.

The cooling process of the ubiquitin–water sphere system can be represented by the heat diffusion equation,

$$\frac{\partial T(\mathbf{r},t)}{\partial t} = \alpha \nabla^2 T(\mathbf{r},t),$$
$$\alpha = \frac{k}{\rho c_p}$$

where $T(\mathbf{r},t)$ is the temperature at position $\mathbf{r}$ and time t, and α is the thermal diffusivity, with k as the thermal conductivity, ρ the mass density, and c_p the specific heat at constant pressure. The Laplacian operator ∇^2 accounts for spatial gradients in temperature across the solvated molecular system.

As the system cools from 310 K, heat transfers from the protein core to the surrounding water, propelled by the temperature difference. In simulations, Langevin dynamics enhance this relaxation by linking the system to an implicit thermal reservoir, simulating the impact of diffusive heat dissipation.

2.6 Temperature Quench Echo

Simulations of temperature quench echo were performed to assess the reversibility of thermal disruptions in ubiquitin contained within a water sphere. Beginning at 310 K, the system (`ubq_ws.psf`, `ubq_ws.pdb`) was rapidly cooled to 100 K for 1000 steps and then warmed back to 310 K for an additional 1000 steps utilizing a Langevin thermostat (damping coefficient 5 ps^{-1}). The simulation was performed in the NVT ensemble using a 2 fs timestep and non-periodic spherical boundary conditions. MSM was utilized for extended-range electrostatics. Velocities and overall energies were observed at every timestep to assess thermal response and evaluate dynamics before and after quenching. The output comprised the `ubq_quench.dcd` trajectory along with the `ubq_quench.log` file for additional analysis.

2.7 Modelling the silica surface

To model the silica (SiO) substrate, we began with the crystalline structure of α -quartz, the most stable polymorph of silica under ambient conditions. A slab was cleaved along the (001) surface, which is widely studied due to its thermodynamic stability and relevance in biomaterial interfaces[21]. The slab was constructed with a thickness of several unit cells to ensure bulk-like behavior in the interior and a realistic representation of the surface. Periodic boundary conditions were applied in the lateral (X and Y) directions, while a vacuum layer of 15–20 Å was added along the Z-axis to isolate the surface and enable interaction studies with adsorbates or biomolecules. The resulting silica slab was structurally optimized and exported in a format suitable for subsequent simulations, such as molecular dynamics or quantum mechanical calculations, to investigate interfacial phenomena including adhesion and surface reactivity.

2.8 Steered Molecular Dynamics (SMD)

Steered Molecular Dynamics (SMD) provides an atomic-level approach for investigating the mechanical properties, unfolding routes, and structural changes of biomolecules[22]. SMD simulations allow the investigation of molecular responses to forces by applying controlled external forces to certain atoms, referred to as SMD atoms, while keeping other atoms fixed. This approach successfully mimics situations faced in single-molecule experiments like atomic force microscopy (AFM), offering insights into processes such as protein unfolding, ligand release, and significant conformational changes (refer to Figure 8).

2.8.1 Constant Velocity Pulling

SMD simulations driven by constant velocity pulling were conducted with NAMD and the CHARMM27 force field to explore the mechanical unfolding of ubiquitin. The system was solubilized in a TIP3P water box and represented using the files `ubq_wb.psf` and `ubq_wb.pdb`. A harmonic spring with a stiffness of 7.0 kcal/mol/$Å^2$ was connected to the C-terminal atom, which was drawn along the z-axis at a speed of 0.1 Å/ps, while the N-terminal atom remained fixed with harmonic constraints. Simulations were performed in the NPT ensemble at 310 K and 1 atm, utilizing Langevin dynamics for temperature control (damping coefficient of 1 ps^{-1}) and the Langevin piston method to maintain pressure. Long-range electrostatics were addressed using the Particle Mesh Ewald (PME) approach, with a grid spacing of about 1.0 Å. Nonbonded interactions used a cutoff distance of 12 Å and a switching function beginning at 10 Å to gradually reduce van der Waals forces. The SHAKE algorithm was used to constraint all bonds containing hydrogen atoms, allowing for a 2 fs integration time step. Periodic boundary conditions were implemented in all three dimensions. Regular intervals produced simulation output, comprising trajectory (.dcd), energy (.log), and restart files, which were subsequently post-processed to derive force–extension profiles and evaluate unfolding dynamics.

2.8.2 Constant Force Pulling

Simulations with constant force pulling were conducted using NAMD and the CHARMM27 force field to study how ubiquitin reacts to prolonged mechanical stress. The system, immersed in a TIP3P water box and characterized by `ubq_wb.psf` and `ubq_wb.pdb`, was exposed to a steady force of 5.0 kcal/mol/Å directed at the C-terminal atom along the z-axis, with the N-terminal atom held in place. Instead of using a harmonic spring for constant velocity pulling, the force was applied directly as outlined in `ubq_cforce.conf`. Simulations were conducted in the NPT ensemble at 310 K and 1 atm, employing Langevin dynamics (damping coefficient 1 ps^{-1}) along with the Langevin piston technique. Electrostatics were handled with PME (grid spacing $\sim$1.0 Å), and a cutoff of 12 Å along with a switching distance of 10 Å was applied for nonbonded interactions. Hydrogen bonds were limited using SHAKE, and a 2 fs time step was employed. Periodic boundary conditions were implemented. The 2 ns simulation produced trajectory (.dcd) and log (.log) files, which were used to analyze force–extension behavior and mechanical work.

2.9 Modelling SpA and Silica environment

The B-domain of *Staphylococcus aureus* protein A (SpA), recognized for its ability to adhere to both biological and synthetic surfaces, was utilized to study protein–silica interactions. The atomic framework of SpA was obtained from the Protein Data Bank (PDB ID: 2JWD) and stored as `spa_ready.pdb` following preprocessing tasks that involved eliminating extraneous chains and aligning along the Z-axis to ensure the binding domain faced the silica surface.

The amorphous silica substrate was created utilizing structural information from earlier research and transformed into a slab appropriate for molecular dynamics (MD) simulations. The hybrid system was simulated with the CHARMM27 force field for the protein and the INTERFACE force field for silica, utilizing interface-specific files (`interface_silica.rtf` and `interface_silica.prm`) that outline atom types, charges, and nonbonded parameters for silicon and oxygen atoms.

The SpA–silica complex was created by placing `spa_ready.pdb` roughly 5 Å above the silica surface to prevent steric clashes while allowing initial nonbonded interactions. The configuration was saved as `merged_ready.pdb`. To investigate the protein's inherent unfolding characteristics without considering substrate influences, SpA was dissolved in a cubic TIP3P water box through VMD's Solvate plugin. This hydrated setup simulates physiological conditions and enables comparison of responses between substrate-bound and free proteins.

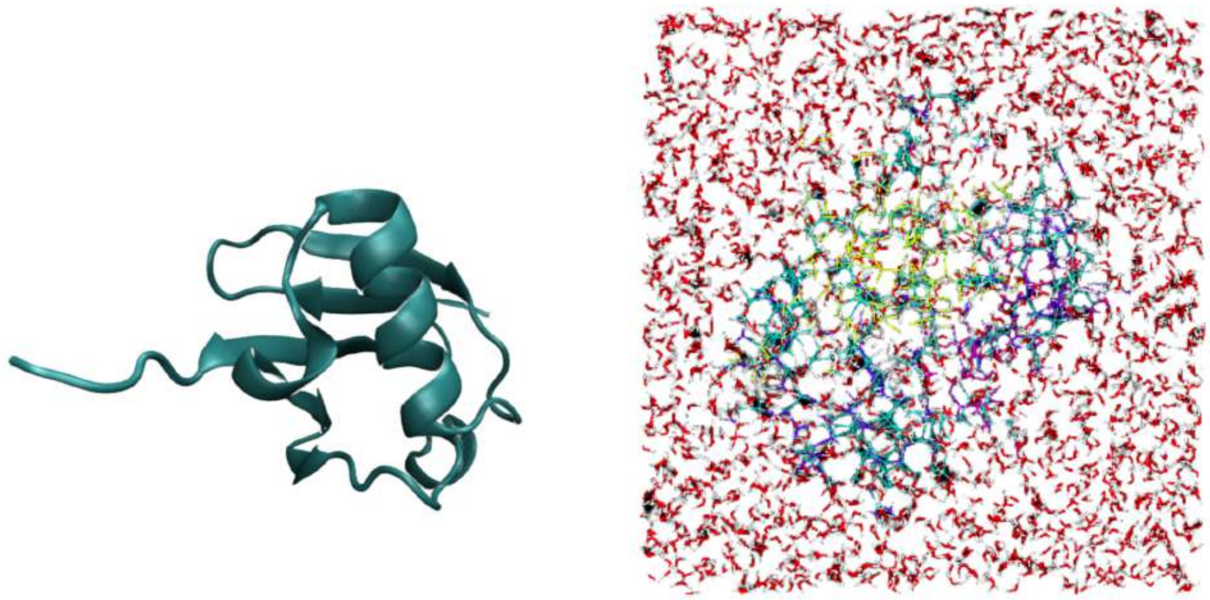
3 Results and Discussion

3.1 System Setup

`ubq.pdb` and `ubq.psf` were obtained after `ubq.pgn` was run on VMD. `ubq.pdb` (Figure 2(a)) file contains the complete coordinates of all atoms, including hydrogens. `ubq.psf` has the complete structural information of the protein.

3.2 Root Mean Square Deviation (RMSD) of Ubiquitin residues

This showed how much each individual residue deviated from the reference position during the simulation. A low RMSD value ($<$ 0.7 Å) suggested a stable secondary structure[23]. Whereas, a high RMSD indicated more flexible residues[10]. The plot for this particular simulation showed a flexible core (0.5 – 1 Å), as opposed to the N and C terminal residues that



(a) (b)

Figure 2: Ubiquitin structures, as visualised on VMD (a) Ubiquitin molecule in 3D ribbon format. (b) Ubiquitin molecule in a water box.

exceeded 1.5 Å. For future simulations, this showed which parts of 1UBQ were the most dynamic (refer to Figure 3).

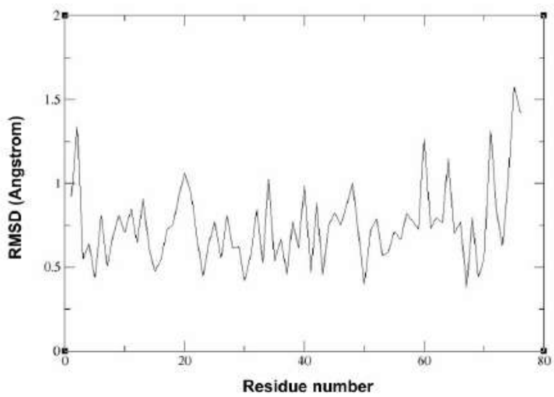


Figure 3: Per-residue RMSD of ubiquitin showing higher flexibility at terminal regions.

A Gaussian Distribution with a bell curve depicts a normal distribution that suggests temperature values that fluctuate around a central mean value with more values near the mean than at the tail ends of the bell[24]. Due to limited number of observations and fewer data points, the curve is not perfectly smooth. However, it does indicate that statistical thermodynamic behavior is successfully followed.

3.3 Cooling Behavior of Ubiquitin in a Water Sphere

The ubiquitin temperature decreases exponentially from approximately 310 K to around 230 K over about 5500 fs, aligning with Newton's law of cooling, demonstrating efficient heat transfer from the protein to the surrounding water. The .log defined for the temperature data of the system was plotted against the simulation time. An exponential decay pattern can be seen which indicates a system dissipating heat and suggests Newton cooling.

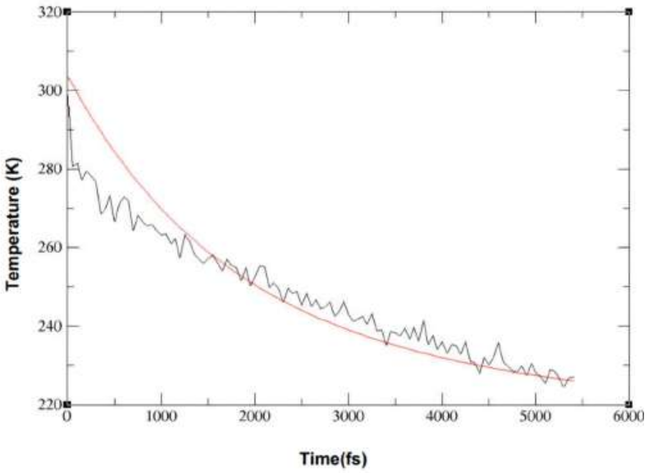


Figure 4: Exponential decay of ubiquitin temperature over time, illustrating Newtonian cooling in a solvated system.

Newton Cooling is the phenomenon where rate of temperature loss is proportional to the temperature difference with the surroundings. The graph indicates more fluctuations at the beginning which reduces with time and shows the energy exchange between ubiquitin and water shell which shows how long it takes the system to reach equilibrium and the kinetics of thermal energy (refer to Figure 4).

3.4 Temperature Quench Echo

An equilibrium simulation is run at T=300K and the coordinate and velocity files hence generated are then run using Xmgrace. The graph is used to indicate coherence, memory effect and reversibility of the system's thermal state. These are temperature echoes which depict the system's return to a prior thermal and retention of phase coherence despite random atomic motions. This reflects the protein's ability to return to an equilibrium state.

The three main regions within the graph exhibit an initial stable plateau at approximately 300 K, signifying equilibrium. A velocity reassignment happens immediately after signifying a strong disturbance and re-equilibration at a lower temperature (refer to Figure 5).

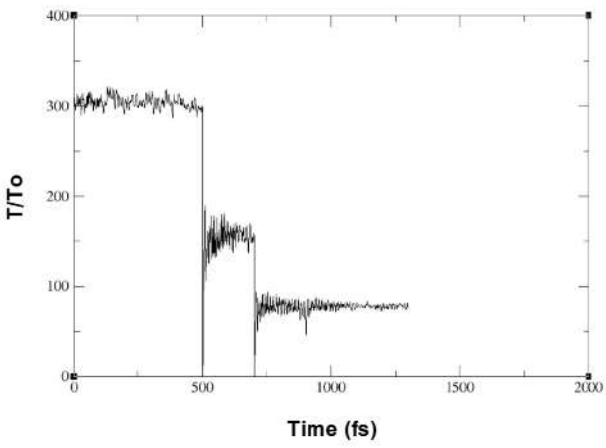


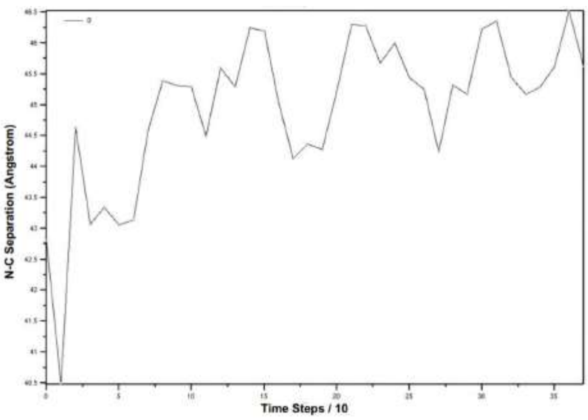
Figure 5: Time evolution of the normalized temperature $\left(\frac{T}{T_0}\right)$ during equilibration.

3.5 SMD : Constant Force Pulling

The N-C terminal separation rises from 40.7 Å to 45.1 Å over 35 time steps, indicating a net elongation of 4.4 Å. The trajectory of ubiquitin under constant force pulling reveals a stepwise increase in the N–C terminal separation, as shown in the first graph. The presence of multiple plateaus in the separation profile suggests that the protein unfolds through a series of stable or metastable intermediate states. Each plateau represents a structural configuration in which the applied force is insufficient to induce further unfolding until a critical threshold is overcome. This behavior indicates that the protein resists deformation until specific intramolecular interactions—such as hydrogen bonds or hydrophobic packing—are disrupted. Such discrete unfolding events are characteristic of globular proteins like ubiquitin, which possess well-defined secondary and tertiary structures (refer to Figure 6).

3.6 SMD : Constant Velocity Pulling

The applied force reaches a high of about 2636 pN near timestep 4000, followed by several local maxima and minima, signifying a sequence of unfolding occurrences. The force decreases to 318.03 pN at timestep 14000, indicating structural relaxation following the initial unfolding. In the constant velocity SMD simulation, the force experienced by the protein exhibits a saw-tooth profile, with a sequence of sharp peaks followed by rapid drops in force. Each peak corresponds to a significant unfolding event, such as the rupture of a β-strand or a domain, requiring a substantial force to overcome. The subsequent force drop reflects a relaxation period as the protein extends and offers less resistance. Over time, the overall trend shows a decrease in peak magnitudes, indicating progressive unfolding and loss of structural integrity[14] . This pattern aligns with previously reported force-extension curves for ubiquitin and supports the view that mechanical unfolding involves sequential domain-level disruptions, each demanding variable energy depending on the structural stability of the segment involved (refer to Figure 7).

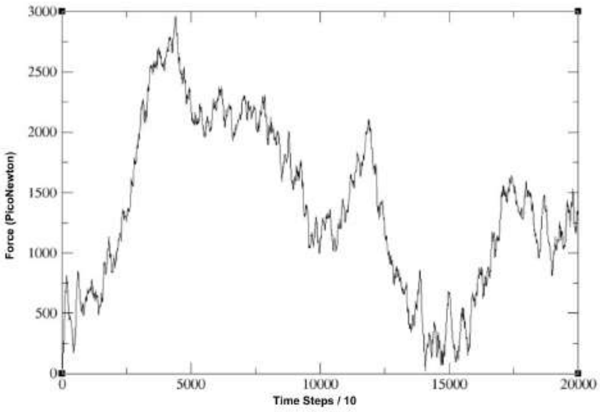


(a)

Timestep / 10	N-C Distance (Å)
0.0	40.2
1.0	44.5
2.0	42.7
3.0	43.0
4.0	43.2
5.0	45.0
6.0	44.8
7.0	45.3
8.0	43.8
9.0	44.9
10.0	45.6
11.0	44.7
12.0	45.0
13.0	44.3
14.0	44.9
15.0	45.9
16.0	44.0
17.0	44.2
18.0	44.7
19.0	45.5
20.0	44.6
21.0	45.1
22.0	45.8
23.0	43.9
24.0	44.1
25.0	43.5
26.0	44.7
27.0	45.6
28.0	45.4
29.0	44.3
30.0	45.7
31.0	44.1
32.0	45.9
33.0	44.0
34.0	46.0
35.0	44.5

(b)

Figure 6: Response of ubiquitin under constant force pulling simulation. (a) Time evolution of N–C terminal separation in ubiquitin. b) Tabulated N–C terminal distances over time.



(a)

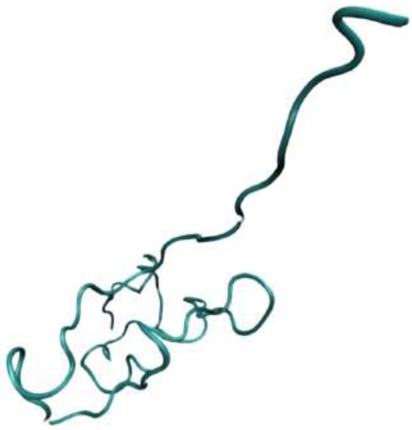
Timestep / 10	Force (pN)
0.0	0.0
1000.0	665.59
2000.0	985.39
3000.0	2007.26
4000.0	2636.27
5000.0	2173.7
6000.0	2256.32
7000.0	2261.11
8000.0	2019.59
9000.0	1588.69
10000.0	1114.73
11000.0	1321.42
12000.0	1863.02
13000.0	811.51
14000.0	318.03
15000.0	656.38
16000.0	546.67
17000.0	1422.08
18000.0	1523.53
19000.0	862.75
20000.0	1348.9

(b)

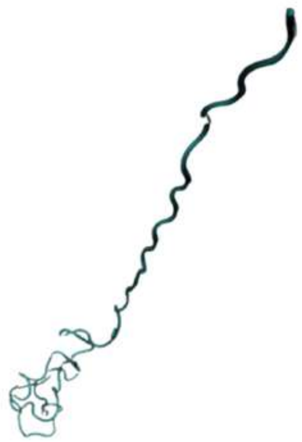
Figure 7: Force response of ubiquitin under constant velocity pulling simulation. (a) Force-extension plot from constant velocity pulling. (b) Extracted data values from force profile.

(a) Structure 1

(b) Structure 2



(c) Structure 3

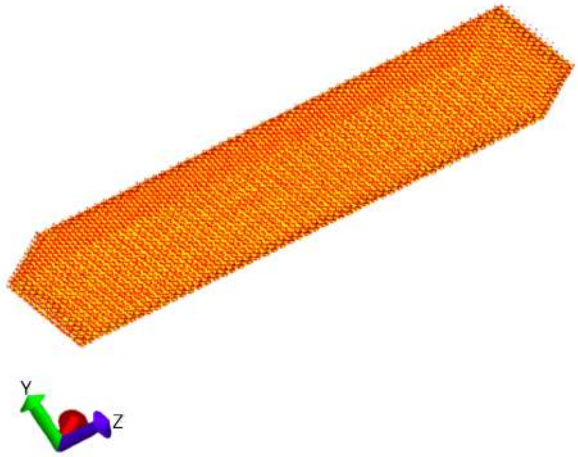


(d) Structure 4

Figure 8: Structures (a)–(d) show consecutive frames from a Steered Molecular Dynamics (SMD) simulation of ubiquitin, carried out using NAMD. The structures were displayed in VMD by extracting frames from the `.dcd` trajectory file. They illustrate how the application of force at the ends triggers gradual unfolding, exposing essential intermediate states and the sequential disappearance of the original secondary structure.

3.7 Silica over VMD

(a)



(b)

Figure 9: Atomistic model of the crystalline silica (α-quartz) supercell used for protein–surface interaction simulations. (a) Side view along the Y–Z plane showing the extended slab thickness (300 Å) to prevent deformation during simulations. (b) Top-down view along the Z-axis illustrating the in-plane periodicity (67 × 69 x 300 Å) of the 14×14×56 supercell, ensuring sufficient surface area to accommodate biomolecular adsorption while minimizing edge effects.

The crystalline silica surface model was constructed using the Atomic Simulation Environment (ASE) library. A trigonal α-quartz SiO unit cell, based on experimentally reported lattice parameters (a = 4.91 Å, c = 5.40 Å), was expanded into a 14×14×56 supercell using a diagonal transformation matrix, yielding approximate dimensions of 67 × 69 × 300 Å. The final system was saved in .xyz and .pdb formats for visualization and simulation setup.

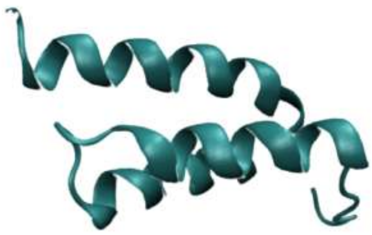
Figure 9(a) shows the side view along the Y–Z plane, where the extended Z-length (300 Å) ensures sufficient slab thickness to prevent substrate deformation during simulations. The atomic structure remains periodic and consistent with the -quartz crystal.

Figure 9(b) provides the top-down view, highlighting the in-plane periodicity across the XYZ plane (67 × 69 x 300 Å), which is suitable for accommodating proteins while minimizing edge and periodic boundary effects.

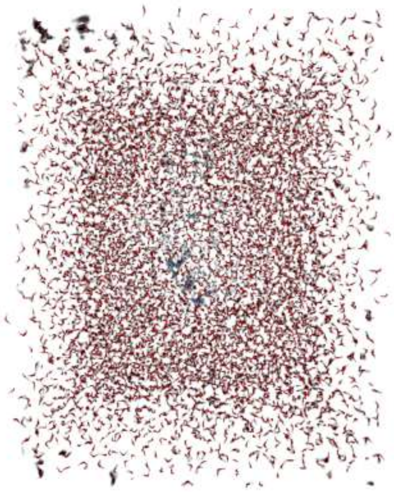
This α-quartz silica supercell serves as a stable, structurally realistic platform for modeling protein–surface interactions, such as Staphylococcus aureus adhesion explored in later sections.

3.8 Modelling Silica and SpA in the same environment

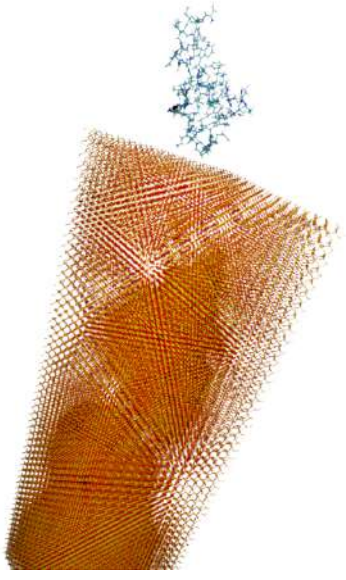
A major challenge in simulating the interaction of *Staphylococcus aureus* protein A (SpA) with silica was the lack of appropriate interface files. Despite the successful preparation and integration of the PDB structure of SpA with the silica slab into a common simulation environment, creating a PSF file with `psfgen` necessitated interface definitions specific to silica. Although CHARMM27 successfully addresses biological elements, simulating inorganic substrates such as silica requires extra interface parameters to accurately represent atomic interactions. Due to the unavailability of these parameters and the constrained project duration, it was not possible to obtain or construct them within the allotted time. As a result, even with a successful structural configuration, the simulation of the SpA–silica system could not be performed (Figure 10).



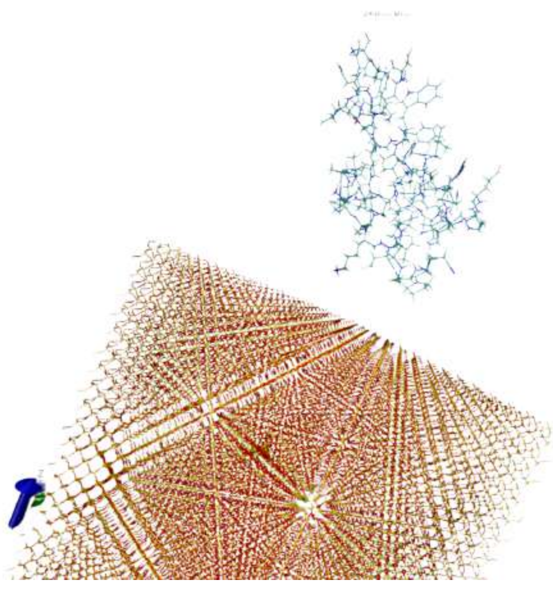
(a) Structure 1



(b) Structure 2



(c) Structure 3



(d) Structure 4

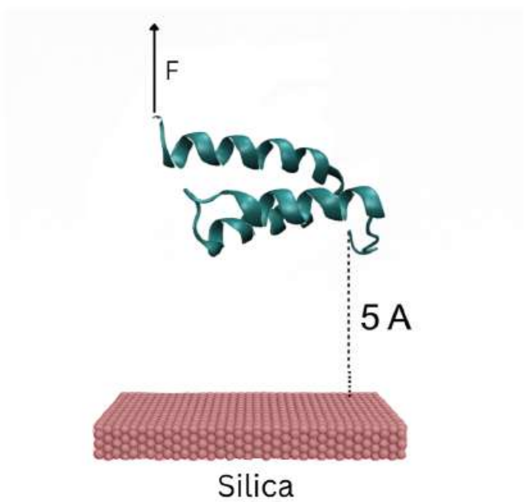
Figure 10: Structural representations of the Staphylococcus aureus surface protein A (SpA) in various simulation environments involving a silica substrate. (a) Isolated SpA protein in its native folded conformation. (b) SpA embedded in a solvated water sphere, illustrating the hydration shell used during the initial equilibration phase. (c) SpA positioned 5 Å above the silica surface, depicting the complete system prepared for protein–substrate interaction simulations. (d) Zoomed-in view of the SpA–silica interface at 5 Å distance, emphasizing molecular-level orientation and potential contact regions relevant to early-stage adhesion behavior.

4 CONCLUSION

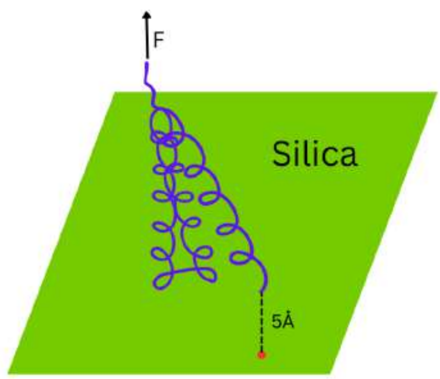
This work presents a molecular dynamics simulation framework developed using ubiquitin as a model protein to streamline system setup, equilibration, and analysis protocols. Simulations were carried out in both explicit and implicit solvent environments using NAMD, with VMD employed for detailed visualization and trajectory analysis. A structurally accurate silica surface was constructed using pymatgen and ASE, providing a robust platform for studying biomolecular interactions at material interfaces. The simulation outcomes closely aligned with established structural and energetic benchmarks, validating the accuracy and consistency of the computational approach.

Molecular dynamics setup and analysis were applied effectively to model adhesin binding to silica, allowing detailed investigation of forces and structural responses at the bio-nano interface.

These findings confirm the protocol's suitability for protein–surface interaction studies and set the stage for the intended exploration of *Staphylococcus aureus* adhesin dynamics on silica surfaces. Insights from these future simulations will deepen our understanding of bacterial adhesion and inform the design of anti-biofilm coatings for medical implants.



(a)



(b)

Figure 11: Force response of ubiquitin under constant velocity pulling simulation. (a) Force-extension plot from constant velocity pulling. (b) Extracted data values from force profile.

A major challenge in simulating the interaction of Staphylococcus aureus protein A (SpA) with silica was the lack of appropriate interface files. Despite the successful preparation and integration of the PDB structure of SpA with the silica slab into a common simulation environment, creating a PSF file with `psfgen` necessitated interface definitions specific to silica. Although CHARMM27 successfully addresses biological elements, simulating inorganic substrates such as silica requires extra interface parameters to accurately represent atomic interactions. Regrettably, these parameters were not readily available, and the short project timeline of two months rendered it unfeasible to acquire or create them independently. As a result, even with a successful structural configuration, the simulation of the SpA–silica system could not be performed (Figure 10).

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